Lonestar6 port: task description for the receiving agent

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What you are being asked to do

Stand up a closed-loop grader-drift experiment on TACC Lonestar6 and run two series of jobs on A100 nodes. The experiment already runs on another cluster (University of Utah CHPC); Lonestar6 is wanted because it has far more GPU capacity. Nothing here is research design — the design is fixed. Your job is the port: build two Python environments, stage the code and data, verify one arm end to end, then launch.

Do not change experiment parameters (rounds, intensities, learning rate, epochs, beta) except the three adaptations listed under "Required changes". Those parameters are measured quantities in a running study and changing them silently invalidates comparisons against runs already finished.

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1. The system you are on

Allocation: `CDA26011`, 50,000 SUs, expires 2027-09-30. Every job must charge it (`#SBATCH -A CDA26011`).

GPU hardware: 84 nodes with 3x NVIDIA A100 PCIe **40 GB** each, 2x AMD EPYC 7763 (128 cores), 256 GB RAM. Also 4 H100 nodes (2x 80 GB) — ignore those unless A100 proves too small.

Queues (from `qlimits`):

queue	max nodes	max wall	max nodes/user	max jobs/user	max submit
`gpu-a100`	8	**48 h**	12	8	32
`gpu-a100-small`	1	48 h	3	3	12
`gpu-a100-dev`	2	2 h	2	1	3
`gpu-h100`	1	48 h	1	1	4

`sinfo` reports GRES as null, which means **nodes are allocated whole, not per GPU**. A job on `gpu-a100` gets all 3 A100s on its node. Our workload is single-GPU, so a naive port wastes two thirds of every node — see "Packing" below.

Storage: `$HOME` 10 GB (too small for anything here), `$WORK` 1 TB (use this), `$SCRATCH` unquota'd but **purged after 10 days of no access** — do not put run outputs there, runs take ~2 days and are analysed for weeks afterwards.

Modules: `python/3.12.11`, `python3/3.9.7`, `cuda/11.3 11.4 12.0 12.2 12.8`. **There is no conda module.** You must install your own Miniconda under `$WORK`.

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2. Getting the code and data

2.1 Code – clone the `tacc-handover` branch

```
git clone -b tacc-handover https://github.com/Zarzard/DriftedGrader.git
```

`main` does NOT contain the experiment: `armorm/` was untracked until this branch. Use the branch.

What you get, which is the transitive import closure of `stage1.py` plus the three launchers:

```
stage1.py              the experiment driver -- everything runs through this
uf_gold.py             loads the UltraFeedback prompt pool
bon_pilot.py           pool format helpers, imported by uf_gold
extract.py             answer extraction
pairmetric.py          pair statistics
verifiers.py           GSM8K answer checking
vintage_score.py       grader scoring helpers
s1gsm12.slurm          the GSM8K launcher (adapt this; it is the newest and cleanest)
s1uf18a.slurm          the UltraFeedback launcher (headline arms)
s1uf18b.slurm          the UltraFeedback launcher (bulk replication)
```

That list was computed, not guessed. If an import still fails, ask for the named file rather than substituting your own – several small modules sit beside these in the source tree.

2.2 Three packages the scoring environment does not pip-install

On CHPC these come from a `pylibs` directory bolted on via `PYTHONPATH`. Do not reproduce that – install them properly:

```
pip install peft==0.13.2 sentencepiece
```

`peft` supplies LoRA and is imported directly by `stage1.py`. `sentencepiece` is needed by the tokenizer, and `stage1.py` asserts it imports at startup. A third, `openai`, also lives in that directory but is imported **inside a function** on the API-judge path, which the loop never takes; `import uf_gold` succeeds without it. Add it only if you later run the API-judge code.

Verify: `python -c "import peft, sentencepiece; print(peft.__version__)"` -> `0.13.2`.

2.3 Data

UltraFeedback prompt pool – in the repo at `data/uf_pool/` (11 MB), nothing to copy:

```
data/uf_pool/uf_armorm.npz              (0.5 MB)
data/uf_pool/uf_armorm_responses.json   (10.4 MB)
```

Both files must stay together with these names: the loader derives the JSON path from the NPZ path by substitution. `uf_gold.py` has a hardcoded `POOL` constant near line 39 still pointing at CHPC. **Edit it to your clone's `data/uf_pool/uf_armorm.npz`.** This is one of the required changes.

GSM8K downloads from HuggingFace automatically: `openai/gsm8k`, config `main`.

Models — four, roughly 60 GB total, all public on HuggingFace:

role	HF id	note
policy	`RLHFlow/LLaMA3-SFT`	8B
grader	`nicolinho/QRM-Llama3.1-8B`	8B, loaded in **fp32**
drifted grader	`nicolinho/QRM-Llama3.1-8B-v2`	8B
gold (UF only)	`RLHFlow/ArmoRM-Llama3-8B-v0.1`	8B

Set `HF_HOME` to a `$WORK` subdirectory before downloading. Pre-download on a login node with `huggingface-cli download <id>`; compute nodes may not have outbound network, so **do not rely on lazy download inside a job**. After downloading, set `HF_HUB_OFFLINE=1` and `HF_DATASETS_OFFLINE=1` in job scripts.

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3. The two environments

Two separate environments are required. They cannot be merged: generation needs a recent vLLM and torch, scoring needs an older torch that the reward-model code depends on.

3.1 `bpref` — scoring, grading, DPO training

```
python 3.8          (3.10 is acceptable if 3.8 is unavailable; test carefully)
torch 2.4.1         CUDA 12.x build
transformers 4.45.2
accelerate 1.0.1
datasets 2.21.0
numpy 1.24.3
scipy 1.10.1
safetensors 0.5.3
peft 0.13.2         (see 2.2)
sentencepiece
```

`stage1.py` asserts `import sentencepiece` succeeds at startup — if it is missing the job exits with a clear message.

3.2 `vllmgen` — generation only

```
python 3.11
torch 2.13.0        CUDA 13.0 build
vllm 0.30.0
transformers 5.17.0
numpy 2.3.5
```

Architecture check. A100 is compute capability **sm_80**, which both environments support natively. Verify after install:

```
python -c "import torch; print(torch.cuda.get_arch_list())"
```

`sm_80` must appear in both. (Note: on CHPC these environments also run fine on sm_89 hardware via PTX JIT, so a missing exact match is not necessarily fatal — but sm_80 should be present outright.)

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4. Required changes (exactly three, plus the queue block)

(a) `gpu_memory_utilization` 0.90 -> 0.80. The experiment runs on 48 GB GPUs elsewhere; the A100 PCIe is **40 GB**. At 0.90 vLLM will try to claim ~43 GB and fail to start. Find the setting in `stage1.py` and lower it. This shrinks the KV cache and will slow generation — that is expected and is what the probe in section 5 measures.

(b) `uf_gold.py` `POOL` path -> `data/uf_pool/uf_armorm.npz` in your clone (see 2.3).

(c) Output directory. The slurm scripts set `D=$Z/graderdrift_runs/stage1` where `$Z` is a CHPC path. Point `$Z` at `$WORK`. Each 12-round arm writes ~1.9 GB of LoRA adapters; a full campaign is ~130 GB, which fits the 1 TB quota. **Not `$SCRATCH`** — it purges at 10 days.

(d) The queue block. Replace the CHPC `#SBATCH` header with Lonestar6's. Note TACC does not use `--gres` here; the partition implies the GPU and nodes come whole.

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5. First task: the timing probe (do this before anything else)

Run **one** GSM8K arm and measure two numbers. Everything downstream depends on them.

Use `gpu-a100-dev` (2 h limit) for a first smoke test that the stack starts at all, then `gpu-a100-small` for a real timed run of 2-3 rounds.

```
stage1.py --task gsm8k --arm stationary \
  --policy RLHFlow/LLaMA3-SFT --policy-step dpo --reward-model none \
  --grader qrm_v1 --grader-drifted qrm_v2 --drift-round 2 \
  --rounds 3 --n-train 3000 --n-eval 500 --n-samples 8 \
  --max-new 400 --batch 8 --epochs 3 --accum 64 --n-shot 8 \
  --dpo-beta 0.6 --gen-backend vllm --out $WORK/runs/probe_stationary
```

Report back:

1. **Hours per round.** On CHPC this is 2.97 h (a6000, UltraFeedback) and 3.90 h (RTX 6000 Ada, GSM8K). A100 should be faster, but the reduced KV cache may offset that. This number decides whether the campaign fits the 48 h wall.
2. **Peak GPU memory**, to confirm 0.80 was the right call and whether it can go higher.
3. **Round-0 held-out accuracy.** Expect **0.76-0.79**. Anything far from that means the model, prompt format, or few-shot setup is wrong — stop and report rather than scaling up.

Sanity values from CHPC for the same configuration: round 0 accuracy 0.758-0.784, rising to ~0.84 by round 3 and holding.

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6. The 48-hour wall is the main obstacle

At CHPC pace:

series	rounds	time needed	fits 48 h?
GSM8K	12	47.7 h	barely, 0.3 h spare
UltraFeedback	18	51.8 h	**no**
UltraFeedback	12	34.9 h	yes

The runs have **no resume by design**. The endogenous drift mechanisms accumulate internal state (`_DRIFT_STATE` in `stage1.py`: a running average `r`, a previous profile `prev`, a cumulative basis `cum`) that no saved adapter reconstructs, so a job killed at the wall loses the entire arm, not just the last round.

Three options, in the order I would try them:

1. **Reduce rounds to fit.** 15 rounds is 43.4 h at CHPC pace. Cheapest, loses the tail.
2. **Make the drift state resumable.** Save `_DRIFT_STATE` next to each round's adapter and reload it with `--resume`. This is a small change but it modifies the thing being measured, so it needs a parity check: one arm run straight through versus the same arm interrupted and resumed must produce identical per-round

numbers.

3. **Split each arm across two chained jobs** with `--dependency=afterok`, which still needs (2).

Decide this with the user after the probe, because if A100 is appreciably faster than the CHPC pace the problem may disappear for 15 or 18 rounds.

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7. Packing: three arms per node

Nodes come whole with 3 A100s. Running one single-GPU arm per job wastes two thirds of the allocation and burns SUs for nothing.

Write the launcher so one job starts **three** arms on its node, each pinned to a different GPU with `CUDA_VISIBLE_DEVICES=0`, `1`, `2`, backgrounded, with a `wait` at the end. Check that three concurrent arms do not contend badly for the 128 cores and 256 GB — each needs ~28 GB of host RAM, so 84 GB of 256 GB, which is comfortable.

With `gpu-a100` allowing 8 concurrent jobs, packed 3-per-node that is **24 arms at once**, against about 6 on CHPC.

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8. What to run once the probe passes

Ask the user which campaign to start; do not assume. The two candidates are:

GSM8K, 7 arms, 12 rounds. Arms: `stationary`, `stationary` (replicate), `matched_random --drift-deg 89`, `matched_random --drift-deg 105`, `momentum --drift-eta 0.50`, `satiation --drift-eta 0.40`, `oracle`. All with `--dpo-beta 0.6`. This is the higher-value series: GSM8K is the only task in the project where gold is a verifiable answer key rather than another model's opinion.

UltraFeedback replication, 46 runs. 23 distinct configurations at two fresh seeds each, to put error bars on arms that currently have none. Needs `--gold-model armorm` and the UF pool from 2.3.

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9. Traps that have already cost time on this project

`--dpo-beta` must be 0.6 on GSM8K. At the old default of 0.1 the policy's mean output length runs away — 664 to 730 to 1946 characters in one round — generations hit the 400-token cap mid-derivation so no answer is emitted, and held-out accuracy collapses from 0.78 to 0.11 at round 3 or 4. Parse rate stays at 1.000 throughout, so the logs look healthy; **length is the only early warning**. Watch `mean_chars` in `history.json` every round. Flat is good; a jump of 2-3x means the run is dead even if accuracy has not fallen yet.

Do not add `--gold-filter`. It was tried, it makes the collapse earlier and deeper, and it is not in the current GSM8K configuration.

`sacct` reporting COMPLETED does not mean the job worked. Two arms once exited in 9-15 seconds with rc=127 because a hand-edited slurm script dropped its environment block, and sacct recorded success. Always confirm a run produced rounds, not just that it exited.

Build new launchers by copying a working one and changing only the arm table. Do not compose one from fragments.

Check output is non-degenerate before trusting a batch. For each finished run verify: `history.json` exists, has the expected number of rounds, `mean_chars` is stable, and `parse_rate` is ~1.0.

10. Definition of done for the port

1. Both environments build, `torch.cuda.get_arch_list()` contains `sm_80`, and `import peft, sentencepiece` succeeds in the scoring environment.
2. Four models and GSM8K are cached under `$WORK`, and a job runs with `HF_HUB_OFFLINE=1`.
3. The probe completes 2-3 rounds with round-0 accuracy in 0.76-0.79.
4. Measured hours per round reported, and the 48-hour decision made with the user.
5. A 3-arms-per-node launcher demonstrated on one node.
6. Campaign launched, with a stated plan for verifying each run's output.

Report the probe numbers before launching anything at scale.